Technological Forecasting & Social Change 209 (2024) 123801

Contents lists available at ScienceDirect

Technological Forecasting & Social Change

journal homepage: www.elsevier.com/locate/techfore

How AI use in organizations contributes to employee competitive advantage: The moderating role of perceived organization support

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A R T I C L E I N F O

Keywords:
Work-related generative AI use
Employee competitive advantage
Employee boundary spanning
Employee agility
Resource base view model

A B S T R A C T

Although the use of generative artificial intelligence (AI) within organizations is becoming increasingly common, research on how to enhance employees' competitive advantage through generative AI use within organizations is very limited. Using a resource-based view model, this study investigates the relationship between generative AI use and employees' competitive advantage, as well as the moderating role of perceived organization support. From an analysis of data from 264 employees from 200 organizations, it is found that work-related generative AI use has a positive effect on employee boundary spanning, which contributes to employee competitive advantage. Secondly, work-related generative AI use also has a positive effect on employee agility, including employee resilience and employee adaptability, which further contributes to employee competitive advantage. However, work-related generative AI use has a positive effect on employee proactivity, while the effect of employee proactivity on employee competitive advantage is not significant. Thirdly, perceived organizational support can enhance the effect between employee boundary spanning and employee competitive advantage. However, it is interesting to observe that perceived organizational support enhances the effect between employee adaptability and employee competitive advantage, while weakening the effect between employee proactivity and employee competitive advantage. It does not exert a moderating effect between employee resilience and employee competitive advantage. These findings can help deepen the current understanding of the relationship between generative AI use in the organization and employee competitive advantage, and provide suggestions for business managers on how to use generative AI to improve employee competitive advantage.

1. Introduction

Recently, generative artificial intelligence (AI) has rapidly gained popularity among various organizations as a tool to enhance efficiency, decision-making, and innovation (Monod et al., 2024). A survey conducted by McKinsey in mid-April 2023 showed that the use of generative AI tools is already very common, and 79 % of the respondents said that they had at least one exposure to generative AI tools at work or outside of work.[1] In July 2023, a survey conducted by Workday covering 1000 global organizations found that over 90 % of organizations currently used AI in their operations to effectively manage personnel, finances, and other aspects. In addition, 80 % of respondents believed that AI helped to improve employee efficiency and decision-making capabilities. The vast majority of people now believe that investing in AI is crucial for maintaining competitiveness.[2] Although AI has been widely applied in organizations and is believed to improve competitiveness, there is currently unclear consensus on how AI can enhance employees' competitive advantage. In view of the increasingly widespread application of AI, how to enhance the employee's competitive advantage become an important question that concerns enterprise managers. Hence, there is a particularly urgent need to explore the relationship between generative AI use within an organization and employee's competitive advantage.

A review of the prior literature shows that the study of competitive advantage can be divided into two major scholarly camps, with a focus on industry-based and resource-based competitive advantage. Researchers in the former camp believe that competitive advantage originates from industrial structure and market environment, rather than

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[1] https://www.mckinsey.com/capabilities/quantumblack/our-insights/the-state-of-ai-in-2023-generative-ais-breakout-year.
[2] https://www.chinaz.com/2023/0710/1541636.shtml.

https://doi.org/10.1016/j.techfore.2024.123801
Received 1 April 2024; Received in revised form 6 August 2024; Accepted 30 September 2024
Available online 4 October 2024

from internal strategies or organizational resources (Porter and Kramer, 2002). In this view, competitive advantage mainly arises from four key factors: production factors, demand conditions, related industries, and corporate strategy (Singh et al., 2019; Thibeault et al., 2023). However, researchers in second camp believe that organizations' competitive advantages come from unique resources and capabilities that are difficult for other organizations to replicate or imitate (Sullivan et al., 2023). These researchers emphasize the core competencies and sustainable competitive advantage of organizations, and believe that organizations should maintain these advantages through continuous learning and adapting to market changes (Zhang et al., 2023).

Following the development of generative AI, some researchers have begun to explore the impact of AI use on employees and organizations (Lee et al., 2023). Existing studies have mainly focused on the adoption of AI applications by employees (Song et al., 2022; Merhi, 2023), the impact of AI on employee's trust (Wong et al., 2023), innovative work behavior (Odugbesan et al., 2023), knowledge management (Arias-Pérez and Vélez-Jaramillo, 2022), and organization performance (Mikalef et al., 2023; Wamba, 2022), as well as the negative impact of AI (Rana et al., 2022). With regard to the relationship between AI use and competitive advantage, most researchers have studied this relationship from the perspective of capability (Mikalef and Gupta, 2021; Hossain et al., 2022; Kemp, 2023). Awamleh and Bustami (2022) proved that AI affects competitive advantage through the mediating role of IT capabilities. However, studies with regard to the relationship between AI use and competitive advantage from a resource-based perspective is rather limited. Among the limited study, Krakowski et al. (2023) found that AI adoption triggers interrelated substitution and complementation dynamics, which can generate new sources of sustained competitive capabilities. However, these researchers did not consider the employees' adaptability or it's effect on competitive advantage. To our knowledge, no researchers have examined how AI use affects employees' adaptability, which further contributes to employee's competitive advantage. Employee adaptability, which includes employee agility and boundary spanning, significantly influences how effectively an employee can utilize digital technologies in their work (Braun et al., 2017). Thus, employee adaptability is an undeniable factor must consider when investigating factors affect employee's competitive advantage. Therefore, different from previous studies, this study investigates how AI use in organizations affects employee competitive advantage from the perspective of employee adaptability.

Another important question relates to the boundary conditions between employee AI use in the organization and employee competitive advantage. According to O'Reilly survey report, 22 % of respondents believed that lack of institutional support was the major obstacle to AI use.[3] However, it is currently unclear what role perceived organizational support plays in the impact of AI use on employee competitive advantage. In particular, this study holds that in the process of employee AI use affecting employee competitive advantage, the relationship between employee adaptability and employee competitive advantage may depend on perceived organizational support. For example, an employee with excellent agility and boundary spanning abilities may have a competitive advantage in their job (Jolly et al., 2021; Baranik et al., 2010). However, if they feel that their organization does not appreciate or acknowledge their abilities, their advantage maybe diminished. In contrast, if an employee perceives strong organizational support, such as opportunities for development, recognition for their contributions, and a positive work environment, their competitive advantage may be enhanced (Lam et al., 2016; Yang and Zhou, 2022). It is therefore important for organizations to recognize the important role of perceived organizational support between employee adaptability and employee competitive advantage. To the best of our knowledge, this is the first study to consider this issue in the context of AI. Unlike prior research, this study aims to examine the moderating effect of perceived organizational support between employee adaptability and employee competitive advantage in the context of AI. Thus, this study investigates the following previously neglected questions:

How does work-related AI use affect employee competitive advantage?

To fill these research gaps, this study presents a research model based on a resource-based view (RBV) to investigate the relationship between work-related AI use and employee competitive advantage, and to explore the relevant boundary conditions. In this way, this study provides theoretical and practical contributions to the literature in the areas of employee AI use and competitive advantage. Firstly, this study represents a contribution to the fields of competitive advantage and RBV models by revealing the relationship between work-related AI use and employee competitive advantage. Secondly, this study contributes to the literature on AI use and employee competitive advantage by indicating how AI use affects employee competitive advantage through the mediation of employee adaptability, including employee boundary spanning and employee agility. Thirdly, this study contributes to the competitive advantage literature by pointing out the boundary conditions of perceived organizational support between employee adaptability and employee competitive advantage in the context of AI. Finally, this study provides practical suggestions for the ways in which AI can be used to improve employee competitive advantage.

2. Theoretical background

2.1. AI use in organizations

The AI mentioned in this study refers to generative AI. Generative AI refers to artificial intelligence that is capable of generating new and original content, such as text, images, audio, and video, based on learned patterns or algorithms (Haenlein and Kaplan, 2019). Compared to other digital technologies, the main technological feature of AI is the ability to cognize and self-learn (Lee et al., 2023; Monod et al., 2024). With its human-like functions, AI not only supports employees' ability to handle complex tasks, but can also replace them through the automation of simple, repetitive tasks (Zirar et al., 2023; Zhang et al., 2024). Consequently, increasing numbers of organizations are applying AI to workflows to increase employee productivity and competitive advantage (Wamba, 2022). Current research on AI use dimensions and organization outcomes in the literature can be seen in Table 1.

Studies have found that AI's augmentation or complementary effects on human-AI interactions have produced positive outcomes for employees, such as resilience, work flexibility, and innovative performance (Chen et al., 2022; Malik et al., 2022; Hu and Pan, 2023). However, researchers have also found negative and unintended consequences of AI use, such as algorithmic discrimination and inequality (Akter et al., 2021; Luengo-Oroz et al., 2021). Existing studies have provided insights into the advantages and disadvantages of AI use that can help organizations to make better use of AI technology, little attention has been paid to how AI can be utilized to develop competitive advantage. AI's capabilities can contribute to develop competitive advantage (Sahoo et al., 2024), every organization can access these digital resources by deploying AI due to the low imitation barriers associated with AI. Therefore, it is essential to explore how AI technology can be combined with other resources to create unique assets that enhance both employee and organizational competitive advantage. The limited research indicates that scholars have highlighted the significance of human cognitive abilities, showing how they complement AI to create sustained competitive advantage (Krakowski et al., 2023). Hence, this study explores how work-related AI use affects employees' competitive advantage.

2.2. Resource-based view model

The RBV provides a theoretical framework for understanding how

[3] https://www.oreilly.com/radar/ai-adoption-in-the-enterprise-2020/.

Table 1
AI use dimensions and organization outcomes in the literature.

Author	Use dimension	Mediator variable	Moderator variable	Dependent variable	Conclusion
Hu and Pan (2023)	Routine use Infusion use	Task-focused, emotion-focused, avoidance coping	N/A	Individual resilience	Infusion use promotes task-focused coping, routine and infusion use promotes emotion-focused coping, routine use inhibits avoidance coping, and task-focused coping and emotion-focused coping promote individual resilience.
Tang et al. (2022)	Artificial intelligence work usage	Role breadth self-efficacy, Role ambiguity	Conscientiousness	Task performance	AI use promotes role breadth self-efficacy and role ambiguity, role breadth self-efficacy promotes task performance, role ambiguity inhibits task performance.
Wang et al. (2022)	Routine use Innovative use	Internal chatbot agility, External chatbot agility	N/A	Customer service performance	Routine use and innovative use promote internal chatbot agility, routine use and innovative use promote external chatbot agility, internal chatbot agility and external chatbot agility promote customer service performance.
Wijayati et al. (2022)	Artificial intelligence	N/A	Change leadership	Work engagement, Employee performance	Artificial intelligence promotes work engagement and employee performance, change leadership positively moderates the influence of AI on employee performance and work engagement.
Zhu and Kanjanamekanant (2022)	Deployment breadth Deployment depth	Work intensification, Job autonomy, Perceived RPA performance	N/A	Burnout Continuance intention to use RPA	Deployment depth promotes work intensification while deployment breadth reduces work intensification, deployment breadth promotes job autonomy, deployment breadth and depth promote perceived RPA performance, work intensification leads to burnout, and job autonomy and perceived RPA performance promote continuous intention.
Cheng et al. (2023)	AI adoption	Challenge appraisals, Hindrance appraisals	Internal locus of control External locus of control	Promotion-focused job crafting, Prevention focused job crafting	AI adoption promotes promotion-focused job crafting and prevention-focused job crafting.
Dutta et al. (2023)	AI-enabled Chatbots	Climate for trust	Past performance, Age	Employee engagement	AI use promotes employee engagement, AI use and employee engagement is mediated through the climate for trust.
Jia et al. (2023)	AI-human collaboration	Answering customers' questions, job design, positive and negative emotions	Employee's job skills	Employee creativity	AI assistance in generating sales leads, on average, increases employees' creativity in answering customers' questions during subsequent sales persuasion. However, this effect is much more pronounced for higher-skilled employees. AI assistance changes job design by intensifying employees' interactions with more serious customers.
Kong et al. (2023)	Perceived AI-supported autonomy	Exploration	AI trust, Proactive personality	Employees' innovative performance	Perceived AI-supported autonomy is positively related to innovation via exploration. This mediation is moderated by AI trust and proactive personality.
Prentice et al. (2023)	AI performance	Employee job engagement, service performance	Job security	Employee job performance	AI performance had a significant effect on job engagement, and employee service performance, which were significantly related to job performance appraisal. The moderation effect exerted by job security was significant in enhancing employees' job engagement and service performance.
Shaikh et al. (2023)	AI	Knowledge sharing, Employee mental health and well-being	Technology leadership	Employee productivity	AI promotes employee productivity, AI promotes knowledge sharing and employee mental health and well-being, knowledge sharing and employee mental health and well-being promote employee productivity.
Tang et al. (2023)	Interaction frequency	Need for affiliation, Loneliness	Attachment anxiety	Help behavior, After work insomnia, After work alcohol consumption	Interaction frequency promotes need for affiliation and loneliness, need for affiliation promotes helping behaviors, loneliness promotes alcohol consumption.
Yin et al. (2024)	AI-assistant	Creative self-efficacy, STARA (smart technology, artificial intelligence, robotics, and algorithms) awareness	Organizational AI readiness	AI-enabled innovation behavior	AI-assistant promotes creative self-efficacy, creative self-efficacy promotes AI-enable innovation behavior, STARA awareness inhibits AI-enable innovation behavior.

organizations manage their physical, human, and organizational resources to gain competitive advantage. The researchers of the RBV argue that organizations derive their sustainable competitive advantage from the value, imperfectly imitable, and irreplaceable resources (Kraaijenbrink et al., 2010). In other words, organizations need to integrate resources to form heterogeneous and immobile resources to maintain their competitive advantage (Pereira and Bamel, 2021). Following the development of RBV, the creation of unique and valuable employee resources has become the focus of researchers in this field (Collins, 2021). Many researchers believe that competent employees are the foundation of an organization's competitive advantage (Narayanan and Nadarajah, 2022; Shet, 2024). Competitors can diminish an organization's competitive advantage by introducing the same technology, and by copying their products and other core resources, but it is difficult for them to attract and imitate competent employees (Elbaz et al., 2018). Employee competitive advantage means that employees not only have the unique skills needed to achieve high performance and innovation but can also quickly adapt and respond to external changes (Kamprath and Mietzner, 2015). Hence, researchers believe that the development of competent employees is the key to the formation of an organization's unique human capital (Van Esch et al., 2018). Current research has focused on the impact of human resource development and training on employee competency (Van Esch et al., 2018), but in recent years, as AI has penetrated into business management, researchers have begun to emphasize AI-driven competitive advantage (Kemp, 2023). AI can change the sources of competitive advantage by substituting or complementing influences on human cognitive abilities (Krakowski et al., 2023). However, it is unclear how AI use affects employee competitive advantage. Using an RBV model, this study therefore explores how the technological resources of AI can be combined with human and organizational resources to produce a unique employee competitive advantage.

3. Hypothesis development

3.1. AI use and employee boundary spanning

Boundary spanning refers to obtaining external resources through action and communication, and includes three main types of external activities: representation, coordination of task performance, and general information search (Ancona and Caldwell, 1992; Van Osch and Steinfeld, 2016). The development of information technology has allowed every employee to become a boundary spanner to access enterprise resources (Zhang et al., 2022). However, boundary spanning as an extra-role behavior leads to role overload, and increases competency demands (Fang et al., 2021). Hence, this study argues that work-related AI use positively affects employees' boundary spanning by removing obstacles to boundary spanning and enhancing boundary spanning capabilities. The reasons are as follows. Firstly, AI can help employees automate repetitive and routine tasks (Jaiswal et al., 2022), which allows them to have the energy and attention to cope with pressures from within and outside the team during boundary spanning. Secondly, when conducting representation and coordination activities with other departments, AI can provide personalized information by analyzing relevant data from each department, thereby improving work efficiency and effectiveness. Finally, AI can provide more accurate feedback to employees by analyzing and self-learning data from employees and the organization (Tong et al., 2021), which can help employees to locate the required information in other departments. Based on the above discussion and the RBV, the following hypothesis is proposed:

H1. Work-related AI use in an organization has a positive effect on employee boundary spanning.

3.2. AI use and employee agility

Employee agility refers to the capacity to adjust quickly and adaptively to internal and external changes, and to use such changes as an opportunity for improvement (Talwar et al., 2023; Alavi et al., 2014). Employee agility can be seen as consisting of the following three main characteristics: proactivity (employees showing anticipatory, spontaneous behaviors to adapt to changing environments), resilience (employees being able to cope with crises and function effectively under pressure), and adaptability (employees being able to change or modify themselves or their behavior to fit better into the new environment) (Pitafi et al., 2018; Cai et al., 2018; Hu et al., 2023). Employee agility is important to allow organizations to adapt and respond to market changes, which can enhance organizational performance and survival in a competitive market environment (Pitafi et al., 2020b). Hence, researchers have focused on how digital technologies such as enterprise social media can enhance employee agility in the digital age (Zhu et al., 2021; Pitafi, 2024). However, it is still unclear how work-related AI use affects employee agility.

Employee agility can be viewed as proactive and adaptive behaviors and a readiness to rapidly redeploy resources (Talwar et al., 2023; Baham and Hirschheim, 2022). Using an RBV model, this study argues that work-related AI use promotes employee agility primarily by providing resources and improving the ability to utilize resources. The reasons may be as follows. Firstly, the cognitive and learning capabilities of AI enable employees to acquire and preserve more instrumental and emotional resources, which provide resources to support proactive behaviors. For example, AI can help employees to obtain large amounts of information from different sources, and can automatically process the information according to the relevant requirements to produce accurate and reliable solutions (Mikalef et al., 2023), which increases employees' decision-making ability and enables them to cope with uncertain situations. Secondly, studies have found that AI can provide mental health support to employees to increase positive emotions (Hu and Pan, 2023), which can increase employee resilience. For example, AI can analyze work data to detect indicators of employee stress, and can provide suggestions for coping. Thirdly, AI can provide feedback to provide developmental direction and guide employees to allocate resources (Tong et al., 2021). For example, employees using AI's identification and problem-solving capabilities can accurately dedicate resources to key elements of problem solving, which improves employee adaptability (Verma and Singh, 2022). Finally, AI can provide feedback on optimizing work processes by analyzing employee-related data (Wang et al., 2022), which can help employees explore new opportunities for improvement. Based on the above discussion and the framework of RBV, this study proposes the following hypotheses:

H2a. Work-related AI use in organization has a positive effect on employee resilience.

H2b. Work-related AI use in organization has a positive effect on employee adaptability.

H2c. Work-related AI use in organization has a positive effect on employee proactivity.

3.3. Boundary spanning, employee agility and competitive advantage

In recent years, researchers have observed that boundary-spanning activities supported by digital technologies such as AI and enterprise social media strongly contribute to employee performance (Prentice et al., 2023; Zhang et al., 2022). This study argues that AI-supported boundary spanning positively impacts competitive advantage by supporting the availability of resources for employees. Boundary spanning allows employees to connect with other teams and departments within the organization to enrich social networks (Van Osch and Steinfeld, 2018), which increases the employee's social capital. Meanwhile, through AI-supported boundary-spanning activities, employees can accurately and quickly identify and acquire heterogeneous knowledge, which updates and expands their knowledge. The knowledge and relational capital that employees obtain through boundary spanning is not only the source of competitive advantage (Díaz-Fernández et al., 2013), but also increases their influence within the team (Liu et al., 2018). As it involves identifying and acquiring useful resources from external and uncertain environments, boundary spanning also exercises employees' abstract thinking and critical skills (Ma et al., 2024). Thus, boundary spanning contributes to employee competitive advantage by providing various resources. Based on the above discussion, this study can propose the following hypothesis:

H3. Boundary spanning has a positive effect on competitive advantage.

According to Pitafi et al. (2018) and Cai et al. (2018), employee agility contains three dimensions including: proactivity, resilience, and adaptability. In particular, proactivity involves leading actions to positively impact one's surroundings, adaptability refers to adjusting behavior in response to new environments for a better fit, and resilience signifies maintaining effective functioning amid stress and changing circumstances, including setbacks in problem-solving. (Hu et al., 2023). Agile employees have the ability to quickly gather and organize relevant information and to effectively use these resources to take appropriate actions (Rasheed et al., 2023). Previous research has pointed out that employee agility helps employees to achieve competitive advantage in dynamic environments (Cai et al., 2018; Talwar et al., 2023).

This study argues that employee agility enables employees to increase their competitive advantage by using AI to access more resources and make more effective resource allocations. The reasons may be as follows. Firstly, facing evolving AI technologies and the dynamic changes they bring to the workplace, agile employees with high resilience can respond quickly to these changes and uncertainties compared to other employees (Ma et al., 2024). Moreover, employee resilience allows employees to keep learning and updating their skills in the dynamic environment (Kuntz et al., 2017), which contributes to individual competitiveness. Secondly, agile employees with high adaptability can better trade-offs between exploratory and exploitative human-AI collaboration modes (Cai et al., 2018), with the aim of accessing the appropriate resources to accomplish specific goals. For example, employee adaptability can fully realize the benefits of AI by proactively exploring innovative uses of AI on the job to identify opportunities while leveraging known AI capabilities to improve productivity (Wang et al., 2022). Third, agile employees with high proactivity can proactively identify new opportunities from AI adoption and proactively adjust work practices to grasp opportunities to enhance individual competitiveness (Krakowski et al., 2023). In short, agile employees, with their strengths in the areas of resource acquisition and utilization, are able to leverage AI to acquire the knowledge, skills, and capabilities needed to maintain a competitive advantage. Based on the above discussion and the RBV, this study proposes the following hypotheses:

H4a. Employee resilience has a positive effect on employee competitive advantage.

H4b. Employee adaptability has a positive effect on employee competitive advantage.

H4c. Employee proactivity has a positive effect on employee competitive advantage.

3.4. Moderating role of perceived organization support

Perceived organizational support refers to employees' beliefs concerning the extent to which the organization values their contribution and cares about their well-being (Riggle et al., 2009). Perceived organizational support is associated with an exchange relationship between the employee and the organization which causes the employee to feel that his or her actions are supported by the resources provided by the organization (Baranik et al., 2010). Employees will then reciprocate towards the organization by creating positive work outcomes. For example, researchers have found that perceived organizational support promotes employees' organizational commitment, work engagement, and so on (Kurtessis et al., 2017). Previous work has also demonstrated the effect of development and training provided by the organization on employee competency (Sung and Choi, 2018). However, it is still unclear how different levels of perceived organizational support impact on the process of AI affecting employee adaptability, such as employee boundary spanning and employee agility, which in turn affects employee competitive advantage.

According to RBV, competitive advantage comes from the integration of resources (Pereira and Bamel, 2021). This study argues that perceived organizational support in the form of organization-based resources can help employees to develop competitive advantage by influencing their behaviors and attitudes. A high level of perceived organizational support means that employees believe that their behaviors are valued by the organization and supported by resources (Kurtessis et al., 2017). Boundary spanning, as an extra-role behavior, not only requires advanced resources, but is also hindered by internal obstacles within the team (Mell et al., 2022). Thus, perceived organizational support gives an employee the resources and motivation to engage in boundary crossing. Based on the above discussion, this study proposes the following hypothesis:

H5. Perceived organization support plays a positive moderating role between employee boundary spanning and employee competitive advantage.

Research has also shown that perceived organizational support relieves employee stress and increases psychological well-being (Kurtessis et al., 2017). Maden-Eyiusta et al. (2022) found that perceived organizational support promotes adaptive behaviors through psychological empowerment. Hence, perceived organizational support allows agile employees to take full advantage of the technological resources provided by AI to complement their capabilities. Conversely, a low level of perceived organizational support means that employees feel that they and their actions are not treated fairly by the organization. As a result, they are less likely to initiate adaptive behaviors, which can reduce their competitive advantage. Based on the above discussion, this study proposes the following hypotheses:

H6a. Perceived organization support plays a positive moderating role between employee resilience and competitive advantage.

H6b. Perceived organization support plays a positive moderating role between employee adaptability and competitive advantage.

H6c. Perceived organization support plays a positive moderating role between employee proactivity and competitive advantage.

This study proposes an RBV research model to investigate the relationship between work-related AI use and competitive advantage, as shown in Fig. 1.

4. Research methods

4.1. Measures

In this study, a questionnaire was used to collect data. Each measurement item was drawn from the literature (see Table 2). Work-related AI use was measured with four items adapted from those used by Ma et al. (2020). Employee competitive advantage was measured with five items revised from Singh et al. (2019). Boundary spanning was measured with four items, which were revised from Faraj and Yan (2009) and Zhang et al. (2022). Employee agility, consisting of employee proactivity, employee adaptability and employee resilience, was measured with items adapted from Pitafi et al. (2018). Employee proactivity was measured with three items, employee adaptability with four items, and employee resilience with three items. Finally, perceived organizational support was measured with four items, which were revised from Kurtessis et al. (2017). All of the above items were scored on a seven-point Likert scale, ranging from one (strongly disagree/unlikely) to seven (strongly agree/likely). In order to ensure the accuracy of translation, three experts in the field of information systems whose native language was Chinese were invited to revise the difficult statements in the questionnaire items. To ensure the reliability and validity of the items in the survey, a pilot test is conducted in this study. The result of the pilot test showed that the measurement items were adequate for further implementation.

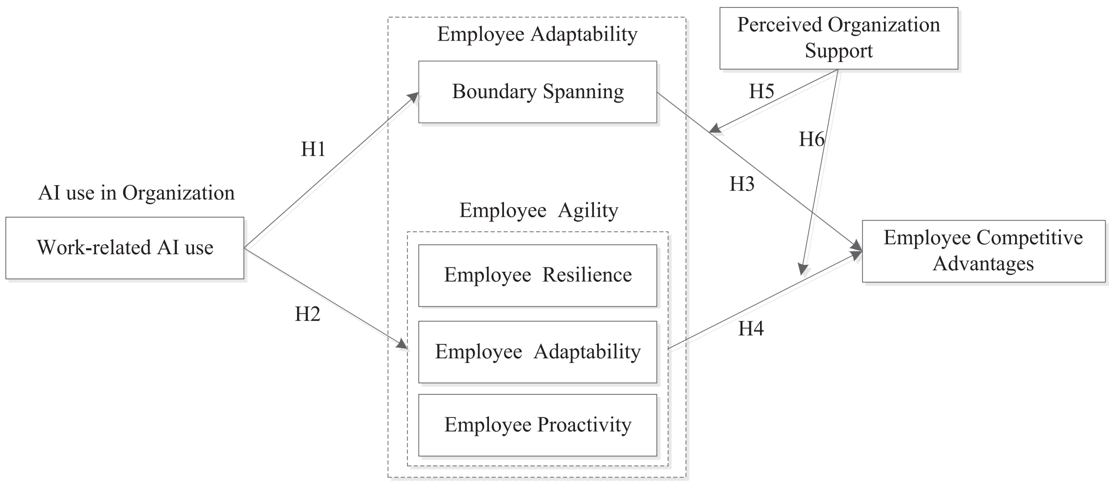

Fig. 1. Research model.

4.2. Data collection

The respondents to this survey were limited to employees within organizations who used AI internally. The AI in this context refers to generative AI, meaning technology that uses machine learning algorithms to learn patterns and features from existing data, and then uses these patterns and features to generate new data with similar features. During the data collection process in this study, AI tools mainly refers to generative AI tools, such as ChatGPT, ERNIE Bot, etc. The questionnaire was administered via an online questionnaire service platform in China, which is similar to the Amazon's MTurk platform and is widely utilized by academics in related fields (Aguinis et al., 2021; Huang et al., 2024). After designing the questionnaire, an electronic questionnaire link was generated and sent to employees from 200 organizations via the platform, to invite them to fill out the questionnaire. Respondents who filled out the questionnaire were given a certain amount of compensation. This study conducted two rounds of data collection, with 320 questionnaires collected in the first round and 164 in the second round. This study collected a total of 484 questionnaires. Among them, we excluded 187 questionnaires from the sample as they did not involve the use of AI in work, leaving us with 297 questionnaires. After removing questionnaires with inconsistent or identical responses, 264 questionnaires remained for analysis in this study. Table 3 shows the descriptive statistics of the respondents.

The organizations in which the respondents worked included those in the information technology and computer service industry (such as Hua Wei, Alibaba, Bai Du, ByteDance, iFLYTEK, Mei Tuan, Star Big Data, State Power Investment Group Digital Technology, Guizhou Big Data Center Research Institute), the manufacturing industry (such as Haier, Fuyao Glass, Xiaomi Company, Midea Company, BYD Company, Dingrun Intelligent Equipment Manufacturing Co., Ltd.), the communication industry (such as Tencent, China Mobile Corporation), and education and culture (such as Zhonggong Education Technology Co., Ltd., Guangzhou Xueerhao Education Consulting Co., Ltd., Peking University, and Guangzhou University). The job titles of the respondents working in the field of technology included engineers, programmers, IT professionals, designers, and data analysts. Those in business included entrepreneurs, marketing managers, HR, grass-roots employees, procurement officers, and sales representatives. For those in education, job titles included teachers, education researchers, and education consultants.

5. Data analysis and results

5.1. Measurement model

In this study, Smart PLS 3.0 software was used to evaluate the measurement model and to test the reliability, convergent validity, and discriminative validity of the data. Cronbach's alpha, the composite reliability (CR), and the average extracted variance (AVE) were used as measures of reliability. As shown in Table 4, the values of Cronbach's alpha ranged from 0.710 to 0.830, and exceeded the threshold value of 0.7; the CR values ranged from 0.837 to 0.885, exceeding the threshold value of 0.7; and the AVE values ranged from 0.587 to 0.665, exceeding the threshold value of 0.5, indicating that all indicators had good reliability. The convergent validity and discriminant validity were measured using factor loadings and the square root of the AVE, respectively. As shown in the results, all factor loadings were >0.7 at a significance level of $p < 0.01$ (see Table 5), indicating good convergent validity, and the square root of the AVE for each factor was greater than its correlation coefficient with other factors (see Table 4), thus confirming discriminant validity.

Given that single-source and cross-sectional data are prone to common method variance (CMV), Harman's single-factor test was used to evaluate the 27 scale items in our model to ensure that the dataset was free of CMV. The results showed that the highest covariance explained by a solitary factor was 39.457 %, which fell below the cut-off value of 50 % (Ma et al., 2020). A collinearity evaluation approach was also used to examine the common method bias, and it was found that the variance inflation factor (VIF) ranged from 1.273 to 2.137, with values below the threshold of 3.3. The method of Liang et al. (2007) was applied to examine common method bias. As shown in Table 6, the results showed that the average variance explained by the substantive indicators was 0.622, in contrast to a much lower average method-based variance of 0.008. Furthermore, most of the factor loadings related to the method were statistically insignificant. Hence, common method bias was not a serious concern in this study. The standardized root mean squared residual (SRMR) was used to assess the goodness of fit for the saturated model. The SRMR value was found to be 0.072, i.e., lower than the suggested value of 0.08, signifying that the model could be considered to have a good fit (Ma et al., 2021). The proposed model therefore displayed a good structural fit with the data.

5.2. Structural model

The results of our analysis of the structural model are shown in Fig. 2. Firstly, work-related AI use has a positive effect on employee boundary

Table 2
Measurement items.

Constructs	Items
Work-related AI use (AW) (Ma et al., 2020)	I use AI tools to obtain ideas and participate in work-related discussions. I use AI tools to acquire solutions to work problems. I use AI tools in my daily work to ask questions. I use AI tools in my daily work to obtain knowledge.
Competitive advantage (CA) (Singh et al., 2019)	My products/services are better than those of my competitors. My R&D capabilities are better than those of my competitors. My managerial capabilities are better than those of my competitors. My profitability is better than that of my competitors. My competitive advantage is better than that of my competitors.
Boundary spanning (BS) (Faraj and Yan, 2009, Zhang et al., 2022)	I use AI tools to solicit information and resources from external channels. I make use of external resources through AI tools to support my work. I actively use AI tools to solicit external work resources. My work depends upon information and resources actively solicited through AI tools, that is, information and resources beyond those coming through official channels.
Employee proactivity (EP) (Pitafi et al., 2018)	I look for opportunities to make improvements at work. I am trying to find more effective ways to perform my job. At work, I stick to what I am told or required to do.
Employee adaptability (EA) (Pitafi et al., 2018)	At work, I can adjust to new work procedures. At work, I can quickly learn to use new equipment. At work, I can quickly learn to keep up to date. At work, I can quickly switch from one project to another.
Employee resilience (ER) (Pitafi et al., 2018)	I am able to perform my job efficiently in difficult or stressful situations. I am able to work well when faced with a demanding workload or schedule. When a new situation arises, I react by trying to manage the problem.
Perceived organizational support (OS) (Kurtessis et al., 2017)	My organization really cares about my well-being. My organization strongly considers my goals and values. My organization cares about my opinions. My organization is willing to help me if I need a special favor.

Table 3
Descriptive statistics of respondent characteristics.

Demographic variable		Size	%
Gender	Male	108	40.91
	Female	156	59.09
Age	20–30 years old	94	35.61
	31–40 years old	144	54.55
	41–50 years old	20	7.58
	> 50 years old	6	2.27
Education	Senior middle school or below	1	0.38
	Junior college	25	9.47
	Bachelor's degree	205	77.65
	Master's degree	30	11.36
	Doctor degree	3	1.14
Income	≤3000	5	1.89
	3001–5000	40	15.15
	5001–8000	81	30.68
	8001–12,000	76	28.79
	12,001–15,000	39	14.77
	>15,000	23	8.71
The time you have using AI	≤1 year	16	6.06
	1–3 years	92	34.85
	3–5 years	104	39.39
	5–10 years	41	15.53
	>10 years	11	4.17
What artificial intelligence products have you used?	ChatGPT	161	60.98
	Voice Assistant (such as Siri)	204	77.27
	Machine translation	134	50.76
	Chatbot	134	50.76
	Image recognition and monitoring	109	41.29
	Unmanned driving (express transportation, taxis, etc.)	62	23.48
	Intelligent customer service	131	49.62
	AI Intelligent Recommendation System	107	40.53
	AI risk assessment	71	26.89
	AI assisted diagnosis	58	21.97
	AI guidance	34	12.88
Nature of the organization	Manufacturing	86	32.58
	Construction	21	7.95
	Finance	20	7.58
	Information technology and computer service industry	69	26.14
	Communication industry	12	4.55
	Communications and transportation industry	11	4.17
	Medical hygiene	9	3.41
	Educational Culture	22	8.33
	Catering service industry	6	2.27
	Coal industry	1	0.38
	Others	7	2.65

spanning ($\beta = 0.688$, $P < 0.001$), and thus H1 is supported. Secondly, work-related AI use has a positive effect on employee resilience ($\beta = 0.498$, $P < 0.001$), employee proactivity ($\beta = 0.563$, $P < 0.001$), and employee adaptability ($\beta = 0.605$, $P < 0.001$), meaning that H2a, H2b, and H2c are supported. Furthermore, employee boundary spanning has a positive effect on employee competitive advantage ($\beta = 0.171$, $P < 0.01$), and H3 is supported. Employee resilience ($\beta = 0.211$, $P < 0.001$) and employee adaptability ($\beta = 0.198$, $P < 0.01$) have a positive effect on employee competitive advantage, and thereforeH4a and H4b are supported. However, the effect of employee proactivity on employee competitive advantage is not significant ($\beta = 0.081$, $P > 0.05$), and H4c is not supported. The R squared value is also shown for the model.

5.3. Mediating effects

The bootstrap method was implemented with the PROCESS SPSS macro to test the mediating effect of employee boundary spanning between employee work-related AI use and competitive advantage, employee agility and competitive advantage. The approach used here has been widely adopted by researchers (Liang et al., 2021; Zhang et al., 2019). Two methods were applied, the bias-corrected method and the percentile method, and a 95 % confidence level was selected for the confidence intervals. According to Prebensen and Xie (2017), a mediator has an effect if the confidence interval of the indirect effect (as measured by the bias correction or percentile method) does not include zero. Specific types of mediation effects were identified, as follows: (a) when the confidence intervals of both the direct and indirect effects do not include zero, a partial mediation effect is present; (b) when the confidence interval of the indirect effect does not include zero, but the confidence interval of the direct effect does include zero, a full mediation effect is present. Table 7 shows our results.

Several important findings can be observed. Firstly, employee boundary spanning partially mediates the relationship between employee work-related AI use and competitive advantage. This means that employee work-related AI use not only positively impacts employee competitive advantage, but also affects this relationship through

Table 4
Descriptive statistics and inter-construct correlations.

Items	Alpha	rho_A	CR	AVE	AW	BS	CA	EA	EP	ER	OS
AW	0.780	0.780	0.859	0.603	0.776						
BS	0.763	0.773	0.850	0.587	0.688	0.766					
CA	0.828	0.830	0.879	0.593	0.579	0.606	0.770				
EA	0.771	0.773	0.853	0.592	0.605	0.571	0.663	0.770			
EP	0.708	0.710	0.837	0.631	0.563	0.470	0.555	0.646	0.794		
ER	0.746	0.750	0.856	0.665	0.498	0.513	0.657	0.647	0.577	0.815	
OS	0.826	0.829	0.885	0.658	0.454	0.542	0.681	0.551	0.444	0.565	0.811

Note: Work-related AI use (AW); Competitive advantage (CA); Boundary spanning (BS); Employee proactivity (EP); Employee adaptability (EA); Employee resilience (ER); Perceived organizational support (OS).

Table 5
Cross loadings.

Items	AW	BS	CA	EA	EP	ER	OS
AW1	**0.798**	0.524	0.423	0.484	0.380	0.339	0.344
AW2	**0.791**	0.548	0.487	0.440	0.458	0.408	0.320
AW3	**0.758**	0.563	0.467	0.488	0.360	0.456	0.446
AW4	**0.757**	0.500	0.418	0.468	0.547	0.341	0.297
BS1	0.519	**0.702**	0.339	0.425	0.456	0.311	0.292
BS2	0.577	**0.790**	0.477	0.540	0.397	0.443	0.411
BS3	0.576	**0.850**	0.514	0.445	0.395	0.409	0.441
BS4	0.429	**0.713**	0.515	0.332	0.192	0.401	0.509
CA1	0.481	0.559	**0.778**	0.550	0.474	0.554	0.549
CA2	0.442	0.464	**0.750**	0.466	0.446	0.419	0.521
CA3	0.371	0.353	**0.755**	0.482	0.457	0.506	0.561
CA4	0.468	0.504	**0.801**	0.491	0.368	0.492	0.536
CA5	0.463	0.441	**0.765**	0.559	0.388	0.551	0.451
EA1	0.413	0.452	0.531	**0.746**	0.434	0.485	0.421
EA2	0.448	0.404	0.477	**0.784**	0.554	0.505	0.357
EA3	0.459	0.394	0.484	**0.763**	0.497	0.447	0.421
EA4	0.534	0.500	0.545	**0.785**	0.506	0.549	0.488
EP1	0.433	0.357	0.490	0.503	**0.770**	0.539	0.438
EP2	0.485	0.407	0.449	0.529	**0.820**	0.446	0.308
EP3	0.420	0.352	0.375	0.507	**0.791**	0.381	0.307
ER1	0.414	0.475	0.534	0.526	0.396	**0.832**	0.480
ER2	0.406	0.426	0.576	0.547	0.451	**0.860**	0.556
ER3	0.399	0.352	0.493	0.510	0.574	**0.751**	0.336
OS1	0.326	0.431	0.586	0.461	0.338	0.508	**0.810**
OS2	0.365	0.486	0.572	0.444	0.324	0.431	**0.854**
OS3	0.393	0.440	0.520	0.435	0.402	0.426	**0.822**
OS4	0.394	0.398	0.526	0.448	0.385	0.466	**0.756**

Table 6
Common method bias analysis.

Construct	Indicator	Substantive Factor Loading (R1)	$R1^2$	Method Factor Loading (R2)	$R2^2$
Work-related AI use	AW1	0.889***	0.790	−0.105	0.011
	AW2	0.802***	0.643	−0.010	0.000
	AW3	0.656***	0.430	0.123	0.015
	AW4	0.755***	0.570	−0.005	0.000
Boundary Spanning	BS1	0.760***	0.578	−0.062	0.004
	BS2	0.719***	0.517	0.084	0.007
	BS3	0.886***	0.785	−0.046	0.002
	BS4	0.692***	0.479	0.022	0.000
Competitive Advantage	CA1	0.582***	0.339	0.214*	0.046
	CA2	0.782***	0.612	−0.034	0.001
	CA3	0.826***	0.682	−0.080	0.006
	CA4	0.889***	0.790	−0.095	0.009
	CA5	0.772***	0.596	−0.006	0.000
Employee Adaptability	EA1	0.748***	0.560	0.003	0.000
	EA2	0.899***	0.808	−0.125	0.016
	EA3	0.817***	0.667	−0.063	0.004
	EA4	0.618***	0.382	0.182*	0.033
Employee Proactivity	EP1	0.646***	0.417	0.141	0.020
	EP2	0.830***	0.689	−0.013	0.000
	EP3	0.903***	0.815	−0.122*	0.015
Employee Resilience	ER1	0.853***	0.728	−0.023	0.001
	ER2	0.860***	0.740	0.003	0.000
	ER3	0.727***	0.529	0.022	0.000
Perceived Organizational Support	OS1	0.775***	0.601	0.033	0.001
	OS2	0.902***	0.814	−0.059	0.003
	OS3	0.866***	0.750	−0.044	0.002
	OS4	0.691***	0.477	0.079	0.006
Average		0.783	0.622	0.001	0.008

employee boundary spanning. Secondly, employee resilience also partially mediates the relationship between employee work-related AI use and competitive advantage. Again, this indicates that employee work-related AI use not only has a positive impact on employee competitive advantage but also influences this relationship through employee resilience. Thirdly, employee adaptability partially mediates the relationship between employee work-related AI use and competitive advantage, which again suggests that employee work-related AI use not only positively impacts employee competitive advantage but also acts through employee adaptability to influence this relationship. Finally, employee proactivity also partially mediates the relationship between employee work-related AI use and competitive advantage. This finding indicates that employee work-related AI use not only has a positive impact on employee competitive advantage but also acts through employee proactivity to influence this relationship.

5.4. Moderating effects

To investigate the moderating role of perceived organizational support between employee boundary spanning and competitive advantage, employee agility and competitive advantage, the stepwise regression method was employed. The advantages of using stepwise regression to test the moderating effects are as follows. Firstly, it allows for the automatic selection of independent variables, thus reducing the influence of human intervention. Secondly, it avoids the problem of overfitting, which improves the generalization ability of the model. Thirdly, it can increase the prediction accuracy and explanatory power of the model, aligning it more closely with the actual situation (Ma and Zhang, 2023). The moderation effect was therefore tested using the stepwise regression method. Employee boundary spanning was considered in model 1, and perceived organization support was then added to model 1 to create model 2. Finally, model 3 was derived by adding to model 2 the interaction between employee boundary spanning and perceived organization support. The results are presented in Table 8.

It can be seen from the Table 8 that perceived organization support has a positive moderating effect between employee boundary spanning and employee competitive advantage. This moderating effect is illustrated in Fig. 3, where the x-axis represents the level of employee boundary spanning, the y-axis represents the level of employee competitive advantage, and the dashed and full lines indicate the relationship between boundary spanning and employee competitive advantage under conditions of high and low perceived organization support. A slope analysis shows that employee boundary spanning has a greater positive effect on employee competitive advantages with higher perceived organization support. Thus, H5 is supported.

Using the same analysis, the moderating role of perceived organizational support between employee agility and competitive advantage

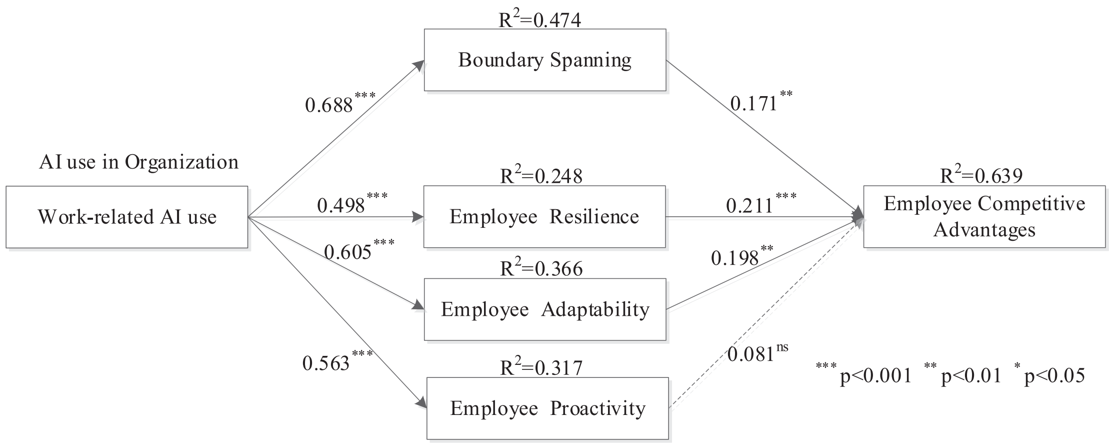

Fig. 2. Results of the structural model analysis.

Table 7
Results of mediating effects.

M/(IV)/(DV)	Items	Effect	Coefficient		Bias-Corrected		Percentile		Mediation existence
			SE	*T*	*95%CI*		*95%CI*		
BS/(AW)/(CA)	Direct effect	0.306	0.064	4.781	0.179	0.432	0.179	0.432	Partial
	Indirect effect	0.259	0.049	5.286	0.164	0.360	0.162	0.359	
ER/(AW)/(CA)	Direct effect	0.327	0.049	6.673	0.231	0.423	0.231	0.423	Partial
	Indirect effect	0.238	0.042	5.667	0.163	0.330	0.156	0.325	
EA/(AW)/(CA)	Direct effect	0.275	0.055	5.000	0.168	0.383	0.168	0.383	Partial
	Indirect effect	0.289	0.050	5.780	0.202	0.397	0.199	0.395	
EP/(AW)/(CA)	Direct effect	0.385	0.056	6.875	0.274	0.496	0.274	0.496	Partial
	Indirect effect	0.179	0.049	3.653	0.092	0.283	0.089	0.281	

Note: Bootstrap 5000 times, IV: independent variable; M: mediator; DV: dependent variable.

Table 8
Moderating role of perceived organization support between employee boundary spanning and employee competitive advantages.

Variable	Model 1		Model 2		Model 3	
	β	T value	β	T value	β	T value
Boundary Spanning (BS)	0.590	12.080	0.325	6.606	0.354	7.001
Perceived Organization Support (POS)			0.493	10.035	0.498	10.212
BS × POS					0.077	2.209
Adjusted R squared	0.355		0.533		0.540	
R squared change	0.358		0.179		0.009	
F change	145.937***		100.706***		4.878*	

was investigated, and the results are presented in Table 9. It can be seen that perceived organization support does not have a positive moderating effect between employee resilience and competitive advantages. Thus, H6a is not supported. One possible explanation is that employee resilience and competitive advantage are complex concepts that may encompass multiple dimensions, and perceived organizational support may only be relevant to some of these dimensions and not others. For example, perceived organizational support may be more strongly related to some aspects of resilience or competitive advantages than others. Another explanation might be that the moderating effect of perceived organizational support may vary depending on the stage of development of the organization. However, it is interesting to observe that perceived organization support has a positive moderating effect between employee adaptability and competitive advantage, while perceived organization support has a negative moderating effect between employee proactivity and competitive advantage. Thus, H6b is supported, but H6c is not supported. The moderating effects are illustrated in Figs. 4 and 5. The results are discussed in the next section.

6. Conclusion

6.1. Key findings

Firstly, the results show that work-related AI use positively affects employee boundary spanning and employee agility. Specifically, following previous research on digital technology and boundary spanning (Van Osch and Steinfeld, 2018), this study finds that AI use promotes employees' boundary spanning behaviors. Although previous research has found that AI can act as a boundary object (Prentice et al., 2023), this study shows that work-related AI use encourages employees to utilize AI for boundary spanning. The effect of work-related AI use on employee agility validates our hypothesis that AI increases employees' resource availability and resource allocation efficiency. Moreover, AI use has the most significant impact on employee adaptability of the three dimensions of employee agility. The reason may be that AI's self-learning and cognitive capabilities provide employees with fast and accurate feedback that is more useful for them to cope with dynamic environments (Tong et al., 2021).

Secondly, the results show differences in the effects of boundary spanning and employee agility on employee competitive advantage. As hypothesized above, boundary spanning has a positive impact on employee competitive advantage. Through boundary spanning, employees acquire unique resources such as heterogeneous knowledge, skill, and social connections, which are considered important sources of competitive advantage (Luthans and Youssef, 2004; Teece, 2014). The resilience and adaptability aspects of employee agility positively affect competitive advantage. However, the effect of employee proactivity on employee competitive advantage is not significant. This may be because

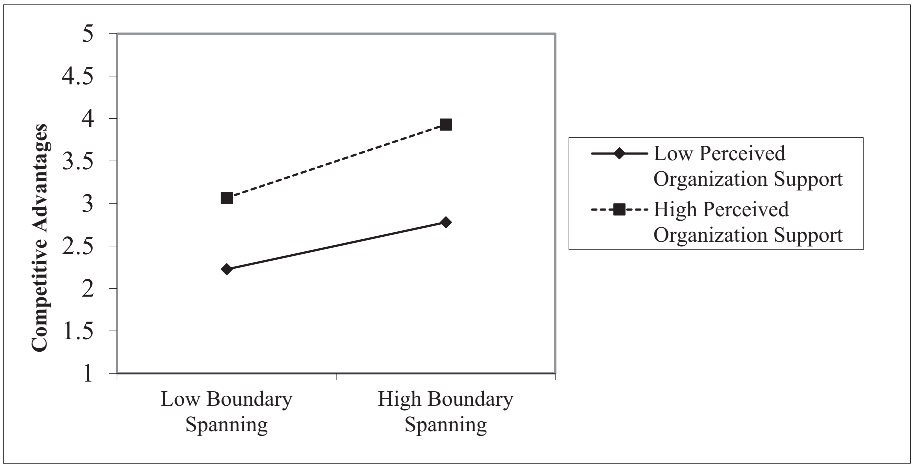

Fig. 3. Moderating role of perceived organization support between employee boundary spanning and employee competitive advantages.

Table 9
Moderating role of perceived organization support between employee agility and employee competitive advantages.

Variable	Model 1		Model 2		Model 3	
	β	T value	β	T value	β	T value
Employee Resilience (ER)	0.354	6.347	0.229	4.312	0.227	4.153
Employee Adaptability (EA)	0.352	5.654	0.247	4.249	0.267	4.562
Employee Proactivity (EP)	0.114	1.976	0.092	1.762	0.046	0.815
Perceived Organization Support (POS)			0.364	7.608	0.376	7.699
ER × POS					−0.002	−0.048
EA × POS					0.162	2.655
EP × POS					−0.138	−2.690
Adjusted R squared	0.527		0.612		0.621	
R squared change	0.532		0.085		0.014	
F change	98.596***		57.887***		3.187*	

although employee proactivity promotes positive work outcomes, proactivity can make employees the target of coworker envy, leading to coworker undermining and a lack of help from coworkers (Sun et al., 2021). Moreover, some corporations require employees to comply with existing practices, which may expose proactive behaviors to resistance and negative consequences.

Finally, the results show that perceived organizational support plays different moderating roles in the relationship between employee adaptability and competitive advantage. Perceived organizational support strengthens the relationship between boundary spanning and competitive advantage, which supports the proposed hypotheses. However, perceived organizational support has different moderating effects on the relationship between the specific dimensions of employee agility and competitive advantage. Perceived organizational support was not found to moderate the relationship between resilience and competitive advantage. The reason for this may be that resilience is relatively stable, and therefore difficult to change (Raetze et al., 2021). Although organizations can foster employee resilience through a series of designed HR practices (Cooke et al., 2019), perceived organizational support may not provide the resources required to foster resilience. Perceived organizational support was found to positively moderate the relationship between adaptability and competitive advantage, which is consistent with previous research that the psychological support provided by perceived organizational support promotes employee adaptability and competence (Kim and Kim, 2022). Perceived organizational support negatively moderated the relationship between proactivity and competitive advantage. This may be because in environments with high levels of organizational support, all employees have access to abundant resources, leaving little room for proactive behaviors to manifest. In

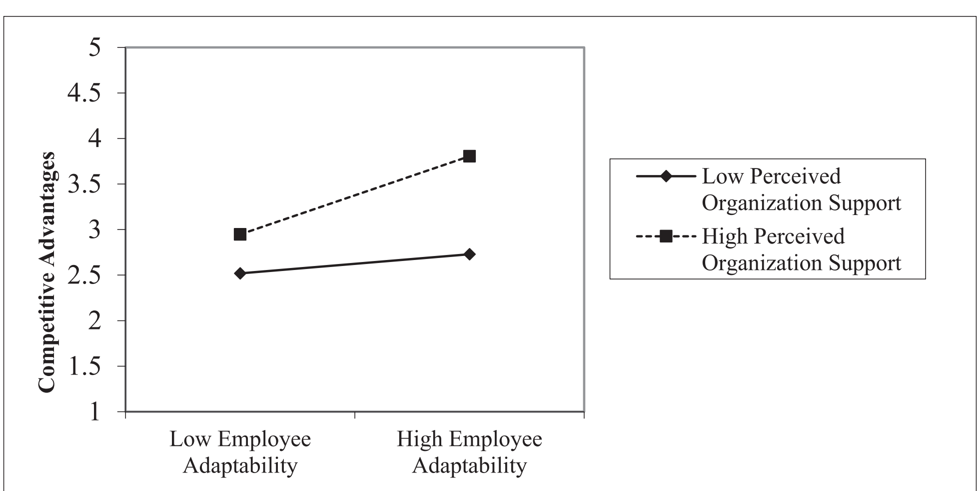

Fig. 4. Moderating role of perceived organization support between employee adaptability and employee competitive advantages.

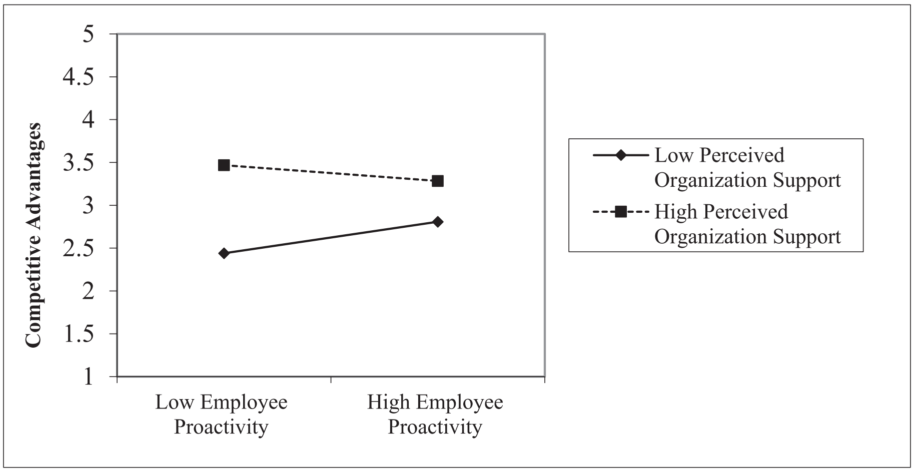

Fig. 5. Moderating role of perceived organization support between employee proactivity and employee competitive advantages.

environments with little organizational support, an employee's proactiveness enables them to create a resource-rich environment for themselves (Wang et al., 2017).

6.2. Theoretical implications

This study gives rise to several theoretical implications. Firstly, this study contributes to the literature on competitive advantage and RBV by revealing the relationship between work-related AI use and employee competitive advantage. Existing research focuses on how the human resource strategies adopted by organizations can develop the basic employee resources needed to sustain competitive advantage (Collins, 2021). However, as a disruptive technology, the potential impact of AI on current industry developments and business operations is enormous. Although some studies have analyzed how AI enhances employee capabilities through complementary and substitution mechanisms (Raisch and Krakowski, 2021), few studies have explored how AI use affects human resources, enabling employees to gain competitive advantage. This study takes a further step by analyzing how the technological resources provided by AI and organizational resources can merge to develop employees' competitive advantage for employees, which enriches the current research on AI use and employees competitive advantage.

Secondly, this study contributes to the literature on AI use and employee's competitive advantage by showing how AI use affects employee competitive advantage through the mediation of employee adaptability, including employee boundary spanning and employee agility. Although scholars have explored the advantages and disadvantages of AI use in organizations, they have focused on the impact on work. For example, AI-enabled task characteristics such as job autonomy and skill diversity have been found to positively promote innovative work behaviors (Verma and Singh, 2022), but AI can also negatively impact job performance by leading to job insecurity and job displacement crises (Tong et al., 2021; He et al., 2023). However, competitors can easily imitate AI-supported technological resources by introducing AI technologies, and few studies have focused on how to merge AI with other resources to create unique resources to sustain competitive advantage. This study extends the current knowledge of this effect by exploring how work-related AI use increases employees' competitive advantage by influencing employee agility and boundary spanning.

Thirdly, this study contributes to the competitive advantage literature by pointing out the boundary conditions of perceived organizational support between employee adaptability and employee competitive advantage in the context of AI. RBV assumes that competitive advantage emanates from the alignment of human resources with organizational resources (Teece, 2014). Although studies have highlighted the impact of the organization's reward and performance appraisal systems on employee competence (Van Esch et al., 2018), it is unclear how the combination of perceived organizational support and employee adaptability affects competitive advantage in the context of AI. This study has filled this gap by finding that perceived organizational support plays different moderating roles in the relationship between employee adaptability and competitive advantage, thereby deepening our understanding of the relationships among employee adaptability, competitive advantage, and perceived organizational support.

6.3. Practical implications

Firstly, this study provides practical implications on how to enhance employees' competitive advantage. Our results show that work-related AI use positively affects competitive advantage through boundary spanning. Hence, organizations need to encourage employees to use AI to access different resources to accomplish their work goals. Organizations also need to develop strategies to encourage resource sharing among departments, to promote boundary-spanning behavior. For example, organizations may require different departments to co-develop or introduce AI technologies to achieve data interoperability. In addition, our results show that employee agility mediates the relationship between AI use and competitive advantage, and that organizations therefore need to focus on developing employee agility. Besides, encouraging employees to use AI to achieve their work goals, organizations can also promote employee agility by introducing AI to facilitate information and knowledge communication among employees (Pitafi et al., 2020a).

Secondly, this study has practical implications in terms of how to provide suitable organizational support to promote competitive advantage when faced with different levels of employee adaptability. Our findings are that perceived organizational support positively moderates the relationship between boundary spanning and competitive advantage. Organizations can therefore encourage and support communication and knowledge sharing among different team members by building an open organizational culture. However, the results show that perceived organizational support exerts different moderating effects on the relationship between the specific dimensions of employee agility and competitive advantage. Hence, organizations need to provide more support to adaptable employees to enable them to achieve competitive advantage. For employees with low levels of proactivity, organizations can compensate for employee resources by creating a supportive organizational culture, but also need to prevent organizational support from reducing the space needed for high-initiative employees to perform well. In addition, organizations need to focus on building resilience through long-term training to increase employee competitive advantage.

6.4. Limitations and future research

Although this study makes contributions to the study of AI use and competitive advantage, it also has certain limitations. Firstly, this study only considered the impact of work-related AI use on competitive advantage, whereas digital technology use with different purposes can have different impacts (Ma et al., 2021). Future research could consider the impact of other purposes of AI use, such as social-related or innovation-related use, on competitive advantage. Secondly, this study used a questionnaire approach, and in future work, case studies could be incorporated to further explore the relationship between AI use and employee competitive advantage. Thirdly, this study only considered boundary spanning and employee agility as mediating mechanisms between AI use and competitive advantage, and the results showed that both of these are only partial mediators. Future research could explore the mediating mechanisms through which AI affects competitive advantage from other perspectives, such as knowledge management. Fourthly, this study only considered the boundary role of organizational support in terms of the organizational dimension, and future research could explore other boundary conditions in regard to the dimensions of leadership and the individual, such as leadership style and digital self-efficacy. Fifthly, although there was no response bias in this study, the age of the respondents was concentrated in young and middle-aged. Considering that there are differences in the acceptance and use of technology among respondents of different ages, future research could nuance the differences in the impact of AI use between younger and older employees. Finally, the subjects in this study were from different industries, which helps to extend the results of this study to various industries. However, considering that there may be differences in what employees do and how AI is specifically applied in different industries, future research could further validate the results of this study in specific industries. For example, future research could explore the differences in the impact of AI use in high-technology and traditional industries based on the degree of technology intensity.

CRediT authorship contribution statement

Liang Ma: Writing – original draft, Conceptualization. **Peng Yu:** Formal analysis, Data curation. **Xin Zhang:** Supervision, Resources. **Gaoshan Wang:** Visualization, Validation. **Feifei Hao:** Writing – review & editing, Software, Methodology.

Data availability

Data will be made available on request.

Acknowledgment

This work was supported by Shandong Provincial Key R&D Plan (Soft Science) under Project No: 2023RZB02021; the General Program of Shandong Natural Science Foundation under Project No: ZR2024MG002; 2023 Higher Education Excellent Youth Innovation Team of Shandong Province under Project No: 2023RW064; 2023 International Cooperation Research Project of Shandong University of Finance and Economics under Project No: 04; and Jinan Municipal School Integration Development Strategy Project "Digital Consumption and Manufacturing Digital Transformation Collaborative Innovation Center Construction" under Project No: JNSX2023052.

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